

ORF 524 – Midterm Exam

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Contents

1 Question 1: Probability Theory (25 points)	1
2 Question 2: Average Risk Optimality (25 points)	3
3 Question 3: Finite-Sample Statistical Inference (50 points)	7

Notes:

- Duration of examination: *4 hours*.
Additional 20 minutes are given to upload your solutions to Canvas.
- All parts are equally weighted (5 points).
Please provide as much detail as possible in your answers.
Answers without proper justification will not receive (partial) credit.
- Please start each question on a separate page.
A single PDF file must be submitted to Canvas.
- Only two pages of personal notes are allowed during the examination.
Basic auxiliary information about standard distributions is provided at the end of the exam.
This is otherwise a closed-book and individual examination.
Do not consult textbooks, online materials, or other sources.
Do not consult with classmates, other people or in online forums.

1 Question 1: Probability Theory (25 points)

Let $(X_1, X_2, \dots, X_n)$ be i.i.d. copies of an integrable random variable $X \sim \mathbb{P}$, with $M(t) = \mathbb{E}[\exp(tX)] < \infty$ for $t \geq 0$. Recall that $X_{(n)} = \max_{1 \leq i \leq n} X_i$, and that parametrizations of basic distributions are given at the end of the exam.

1. Show that

$$\exp(t\mathbb{E}[X_{(n)}]) \leq nM(t), \quad t \geq 0.$$

2. Show that

$$\mathbb{E}[X_{(n)}] \leq \inf_{t>0} \frac{\log n + \log M(t)}{t}.$$

3. Let $\mathbb{P} = \mathcal{N}(0, \sigma^2)$. Show that

$$\mathbb{E}[X_{(n)}] \leq \sqrt{2}\sigma\sqrt{\log n}.$$

4. Let $\mathbb{P} = \text{Gamma}(\alpha, \beta)$. Show that, for $n \geq 2$,

$$\mathbb{E}[X_{(n)}] \leq \frac{2^{1/\alpha}}{2^{1/\alpha} - 1} 2\beta \log n.$$

5. Let $(Z_i : 1 \leq i \leq n)$ be i.i.d. $\mathcal{N}(0, \sigma^2)$ and $(E_i : 1 \leq i \leq n)$ be i.i.d. $\text{Exponential}(0, \theta)$, not necessarily independent. Show that, for $n \geq 2$,

$$\mathbb{E} \left[\max_{1 \leq i \leq n} |Z_i| \sqrt{E_i} \right] \leq \frac{8}{\sqrt{3}} \sigma \sqrt{\theta} \log n.$$

2 Question 2: Average Risk Optimality (25 points)

Let $\mathbf{X} \sim \mathbb{P}_\theta$ be a random vector with $\mathbb{P}_\theta \in \mathcal{P} = \{\mathbb{P}_\theta : \theta \in \Theta\}$. Recall that the risk of a statistic $T(\mathbf{X})$ for the parameter $\Psi(\theta)$ under square loss is $R(\Psi(\theta), T) = \mathbb{E}_\theta[(T(\mathbf{X}) - \Psi(\theta))^2]$.

1. Show that, in general, there is no unique solution (over $\mathcal{P}$) to the optimization problem:

$$\min_{T \in \mathcal{T}} \mathbb{E}_\theta[(T(\mathbf{X}) - \Psi(\theta))^2], \quad \mathcal{T} = \{T(\cdot) : \mathbb{E}_\theta[|T(\mathbf{X})|^2] < \infty\}.$$

2. One way to resolve the lack of uniqueness is to restrict the class of statistics $T(\cdot)$ considered (e.g., allow for unbiased estimators only, that is, restrict to $\mathcal{T}_0$). An alternative solution is to consider *Average Risk*: assume the parameter θ is a random variable with Lebesgue density $f(\theta)$, sometimes called a *prior*, and consider the new optimization problem:

$$\min_{T \in \mathcal{T}_B} \int_{\Theta} \mathbb{E}_\theta[(T(\mathbf{X}) - \Psi(\theta))^2] f(\theta) d\theta, \quad \mathcal{T}_B = \left\{ T(\cdot) : \int_{\Theta} \mathbb{E}_\theta[|T(\mathbf{X})|^2] f(\theta) d\theta < \infty \right\},$$

with $\mathbb{E}_\theta[\cdot]$ now denoting the expectation with respect to the conditional distribution of $\mathbf{X}|\theta$. Show that there exists a *unique* (a.s.) solution to this problem, denoted by $T^*(\mathbf{X})$, and give its precise expression. This optimal procedure is called the *Bayes rule* (for prior $f(\theta)$).

3. Assume $\mathbf{X}$ is a continuous random variable (only for simplicity). Show that

$$T^*(\mathbf{X}) = \frac{\int_{\Theta} \Psi(\theta) f(\mathbf{X}|\theta) f(\theta) d\theta}{\int_{\Theta} f(\mathbf{X}|\theta) f(\theta) d\theta},$$

where $f(\mathbf{X}|\theta)$ denotes the Lebesgue density of the conditional distribution of $\mathbf{X}|\theta$, and $f(\theta|\mathbf{X})$ is interpreted as the Lebesgue density of the (posterior) distribution of $\theta|\mathbf{X}$.

4. Let $\mathbf{X} = (X_1, X_2, \dots, X_n)'$ i.i.d. $N(\theta, \sigma^2)$ conditionally on θ , with $\sigma^2 > 0$ known. In addition, impose a Gaussian prior for θ , that is, let $\theta \sim N(\mu, \tau^2)$, with $\mu \in \mathbb{R}$ and $\tau^2 > 0$.

- (a) Show that $\theta|\mathbf{X} \sim N\left(\mu \cdot \frac{\sigma^2}{n\tau^2 + \sigma^2} + \bar{X}_n \cdot \frac{n\tau^2}{n\tau^2 + \sigma^2}, \frac{\sigma^2\tau^2}{n\tau^2 + \sigma^2}\right)$, where $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$.

Hint: for an invertible matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ and any vector $\mathbf{u} \in \mathbb{R}^n$ such that $\mathbf{u}'\mathbf{A}^{-1}\mathbf{u} \neq -1$,

$$(\mathbf{A} + \mathbf{u}\mathbf{u}')^{-1} = \mathbf{A}^{-1} - \frac{\mathbf{A}^{-1}\mathbf{u}\mathbf{u}'\mathbf{A}^{-1}}{1 + \mathbf{u}'\mathbf{A}^{-1}\mathbf{u}}.$$

- (b) Give the form of the optimal Average Risk (or Bayes) estimator, and compute the associated (Bayes) Risk, when $\Psi(\theta) = \theta$ and when $\Psi(\theta) = \theta^2$.

Discuss what happens when the prior distribution, $f(\theta)$, becomes more concentrated (i.e., $\tau \rightarrow 0$). Explain whether the Bayes procedure for $\Psi(\theta) = \theta$, $\Psi(\theta) = \theta^2$ or other parameters becomes less or more dogmatic (i.e., more dominated by the prior chosen).

3 Question 3: Finite-Sample Statistical Inference (50 points)

Let $(X_i : 1 \leq i \leq n)$ be i.i.d. with $X_i \sim \mathbb{P}_\theta$, $\mathbb{E}_\theta[X_i] = \sqrt{\theta}$, and Lebesgue density

$$f_X(x; \theta) = \frac{x}{a(\theta)} \mathbb{1}(0 \leq x \leq b(\theta)),$$

where $\theta \in \Theta = \mathbb{R}_{++}$, $a, b : \Theta \rightarrow \mathbb{R}_{++}$ are fixed functions, and $\mathbb{1}(\cdot)$ is the indicator function.

1. Consider basic distributional properties of $\mathbb{P}_\theta$.

(a) Find $a(\theta)$ and $b(\theta)$.

(b) Find $F_X(x; \theta) = \mathbb{P}_\theta[X_i \leq x]$, the distribution function of X_i .

2. Consider the (methods of moments) estimator:

$$\hat{\theta}_m = \frac{1}{n} \sum_{i=1}^n m(X_i), \quad \text{for some chosen function } m(\cdot).$$

(a) Let $m(x) = x$. Is $\hat{\theta}_m = \bar{X}_n$ unbiased for θ ? If not, find the direction of the bias.

(b) Find a choice of $m(\cdot)$ such that $\hat{\theta}_m$ is unbiased for θ . Denote this estimator by $\hat{\theta}_U$.

3. Consider the (maximum likelihood) estimator:

$$\hat{\theta}_{\text{ML}} = \frac{4}{9} X_{(n)}^2, \quad X_{(n)} = \max_{1 \leq i \leq n} X_i.$$

(a) Find the distribution of $\hat{\theta}_{\text{ML}}$, and show that $\hat{\theta}_{\text{ML}}$ is a complete sufficient statistics for θ .

(b) Find a UMVU estimator of θ . Denote this estimator by $\hat{\theta}_{\text{UMVU}}$.

(c) Compute $\mathbb{V}[\hat{\theta}_{\text{UMVU}}]$, and show that it is never larger than $\mathbb{V}[\hat{\theta}_U]$. Can these two variances be equal?

4. Consider the following hypothesis testing problem:

$$H_0 : \theta \leq \theta_0 \quad \text{vs.} \quad H_1 : \theta > \theta_0,$$

with a test that rejects H_0 if and only if $\hat{\theta}_{\text{ML}}/\theta_0 > c$ for some critical value c .

(a) Find c so that the test has 5% size.

(b) Derive the power function, denoted by $\beta(\theta)$, and sketch a plot of it.

(c) Propose a $100(1 - \alpha)\%$ (one-sided) confidence interval for θ , where $\alpha \in [0, 1]$.

Table 1.1. Discrete Distributions on $\mathcal{R}$

Uniform	p.d.f.	$1/m, x = a_1, \dots, a_m$
	m.g.f.	$\sum_{j=1}^m e^{a_j t}/m, t \in \mathcal{R}$
$DU(a_1, \dots, a_m)$	Expectation	$\sum_{j=1}^m a_j/m$
	Variance	$\sum_{j=1}^m (a_j - \bar{a})^2/m, \bar{a} = \sum_{j=1}^m a_j/m$
	Parameter	$a_i \in \mathcal{R}, m = 1, 2, \dots$
Binomial	p.d.f.	$\binom{n}{x} p^x (1-p)^{n-x}, x = 0, 1, \dots, n$
	m.g.f.	$(pe^t + 1 - p)^n, t \in \mathcal{R}$
$Bi(p, n)$	Expectation	np
	Variance	$np(1-p)$
	Parameter	$p \in [0, 1], n = 1, 2, \dots$
Poisson	p.d.f.	$\theta^x e^{-\theta}/x!, x = 0, 1, 2, \dots$
	m.g.f.	$e^{\theta(e^t - 1)}, t \in \mathcal{R}$
$P(\theta)$	Expectation	θ
	Variance	θ
	Parameter	$\theta > 0$
Geometric	p.d.f.	$(1-p)^{x-1}p, x = 1, 2, \dots$
	m.g.f.	$pe^t/[1 - (1-p)e^t], t < -\log(1-p)$
$G(p)$	Expectation	$1/p$
	Variance	$(1-p)/p^2$
	Parameter	$p \in [0, 1]$
Hyper-geometric	p.d.f.	$\binom{n}{x} \binom{m}{r-x} / \binom{N}{r}$
		$x = 0, 1, \dots, \min\{r, n\}, r - x \leq m$
$HG(r, n, m)$	m.g.f.	No explicit form
	Expectation	rn/N
	Variance	$rn m(N-r)/[N^2(N-1)]$
	Parameter	$r, n, m = 1, 2, \dots, N = n + m$
Negative binomial	p.d.f.	$\binom{x-1}{r-1} p^r (1-p)^{x-r}, x = r, r+1, \dots$
	m.g.f.	$p^r e^{rt}/[1 - (1-p)e^t]^r, t < -\log(1-p)$
$NB(p, r)$	Expectation	r/p
	Variance	$r(1-p)/p^2$
	Parameter	$p \in [0, 1], r = 1, 2, \dots$
Log-distribution	p.d.f.	$-(\log p)^{-1} x^{-1} (1-p)^x, x = 1, 2, \dots$
	m.g.f.	$\log[1 - (1-p)e^t]/\log p, t \in \mathcal{R}$
$L(p)$	Expectation	$-(1-p)/(p \log p)$
	Variance	$-(1-p)[1 + (1-p)/\log p]/(p^2 \log p)$
	Parameter	$p \in (0, 1)$

All p.d.f.'s are w.r.t. counting measure.

Table 1.2. Distributions on $\mathcal{R}$ with Lebesgue p.d.f.'s

Uniform	p.d.f.	$(b-a)^{-1}I_{(a,b)}(x)$
	m.g.f.	$(e^{bt} - e^{at})/[(b-a)t], \quad t \in \mathcal{R}$
$U(a, b)$	Expectation	$(a+b)/2$
	Variance	$(b-a)^2/12$
	Parameter	$a, b \in \mathcal{R}, \quad a < b$
Normal	p.d.f.	$\frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2}$
	m.g.f.	$e^{\mu t + \sigma^2 t^2/2}, \quad t \in \mathcal{R}$
$N(\mu, \sigma^2)$	Expectation	μ
	Variance	σ^2
	Parameter	$\mu \in \mathcal{R}, \quad \sigma > 0$
Exponential	p.d.f.	$\theta^{-1} e^{-(x-a)/\theta} I_{(a,\infty)}(x)$
	m.g.f.	$e^{at}(1-\theta t)^{-1}, \quad t < \theta^{-1}$
$E(a, \theta)$	Expectation	$\theta + a$
	Variance	θ^2
	Parameter	$\theta > 0, \quad a \in \mathcal{R}$
Chi-square	p.d.f.	$\frac{1}{\Gamma(k/2)2^{k/2}} x^{k/2-1} e^{-x/2} I_{(0,\infty)}(x)$
	m.g.f.	$(1-2t)^{-k/2}, \quad t < 1/2$
χ_k^2	Expectation	k
	Variance	$2k$
	Parameter	$k = 1, 2, \dots$
Gamma	p.d.f.	$\frac{1}{\Gamma(\alpha)\gamma^\alpha} x^{\alpha-1} e^{-x/\gamma} I_{(0,\infty)}(x)$
	m.g.f.	$(1-\gamma t)^{-\alpha}, \quad t < \gamma^{-1}$
$\Gamma(\alpha, \gamma)$	Expectation	$\alpha\gamma$
	Variance	$\alpha\gamma^2$
	Parameter	$\gamma > 0, \quad \alpha > 0$
Beta	p.d.f.	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1} I_{(0,1)}(x)$
	m.g.f.	No explicit form
$B(\alpha, \beta)$	Expectation	$\alpha/(\alpha+\beta)$
	Variance	$\alpha\beta/[(\alpha+\beta+1)(\alpha+\beta)^2]$
	Parameter	$\alpha > 0, \quad \beta > 0$
Cauchy	p.d.f.	$\frac{1}{\pi\sigma} \left[1 + \left(\frac{x-\mu}{\sigma}\right)^2\right]^{-1}$
	ch.f.	$e^{\sqrt{-1}\mu t - \sigma t }$
$C(\mu, \sigma)$	Expectation	Does not exist
	Variance	Does not exist
	Parameter	$\mu \in \mathcal{R}, \quad \sigma > 0$

Table 1.2. (continued)

t-distribution	p.d.f.	$\frac{\Gamma[(n+1)/2]}{\sqrt{n\pi}\Gamma(n/2)} \left(1 + \frac{x^2}{n}\right)^{-(n+1)/2}$
	ch.f.	No explicit form
t_n	Expectation	0, ($n > 1$)
	Variance	$n/(n-2)$, ($n > 2$)
	Parameter	$n = 1, 2, \dots$
F-distribution	p.d.f.	$\frac{n^{n/2}m^{m/2}\Gamma[(n+m)/2]x^{n/2-1}}{\Gamma(n/2)\Gamma(m/2)(m+nx)^{(n+m)/2}}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$F_{n,m}$	Expectation	$m/(m-2)$, ($m > 2$)
	Variance	$2m^2(n+m-2)/[n(m-2)^2(m-4)]$, ($m > 4$)
	Parameter	$n = 1, 2, \dots$, $m = 1, 2, \dots$
Log-normal	p.d.f.	$\frac{1}{\sqrt{2\pi}\sigma}x^{-1}e^{-(\log x - \mu)^2/2\sigma^2}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$LN(\mu, \sigma^2)$	Expectation	$e^{\mu+\sigma^2/2}$
	Variance	$e^{2\mu+\sigma^2}(e^{\sigma^2} - 1)$
	Parameter	$\mu \in \mathcal{R}$, $\sigma > 0$
Weibull	p.d.f.	$\frac{\alpha}{\theta}x^{\alpha-1}e^{-x^\alpha/\theta}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$W(\alpha, \theta)$	Expectation	$\theta^{1/\alpha}\Gamma(\alpha^{-1} + 1)$
	Variance	$\theta^{2/\alpha} \left\{ \Gamma(2\alpha^{-1} + 1) - [\Gamma(\alpha^{-1} + 1)]^2 \right\}$
	Parameter	$\theta > 0$, $\alpha > 0$
Double Exponential	p.d.f.	$\frac{1}{2\theta}e^{- x-\mu /\theta}$
	m.g.f.	$e^{\mu t}/(1 - \theta^2 t^2)$, $ t < \theta^{-1}$
$DE(\mu, \theta)$	Expectation	μ
	Variance	$2\theta^2$
	Parameter	$\mu \in \mathcal{R}$, $\theta > 0$
Pareto	p.d.f.	$\theta a^\theta x^{-(\theta+1)}I_{(a,\infty)}(x)$
	ch.f.	No explicit form
$Pa(a, \theta)$	Expectation	$\theta a/(\theta - 1)$, ($\theta > 1$)
	Variance	$\theta a^2/[(\theta - 1)^2(\theta - 2)]$, ($\theta > 2$)
	Parameter	$\theta > 0$, $a > 0$
Logistic	p.d.f.	$\sigma^{-1}e^{-(x-\mu)/\sigma}/[1 + e^{-(x-\mu)/\sigma}]^2$
	m.g.f.	$e^{\mu t}\Gamma(1 + \sigma t)\Gamma(1 - \sigma t)$, $ t < \sigma^{-1}$
$LG(\mu, \sigma)$	Expectation	μ
	Variance	$\sigma^2\pi^2/3$
	Parameter	$\mu \in \mathcal{R}$, $\sigma > 0$